Representation Theory Seminar Notes

Poisson Geometry and Integrable Systems

Fall 2026

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3 Kirillov-Kostant Poisson Structure on Matrices

Speaker: Vitaly Tarasov (IU Indianapolis)
Date: October 2, 2026

We begin with the Lie-Poisson construction and specialize it to matrix entries. Conjugation flows identify the central functions; invariants of nested corners then supply noncentral commuting Hamiltonians. Tensor notation connects this construction to spectral-parameter identities, developed further in Section 4.1. All polynomial algebras in this section are over $\mathbb{C}$.

3.1 Motivation: recalling the Lie-Poisson bracket

The previous talk associated a commutative Poisson algebra to a Lie algebra. To recall the construction, let $\mathfrak{g}$ have basis $e_1, \dots, e_d$, with

$$[e_i, e_j] = \sum_{k=1}^{d} c_{ij}^k e_k.$$

Every $a \in \mathfrak{g}$ defines a linear function on the dual space:

$$\ell_a : \mathfrak{g}^* \longrightarrow \mathbb{C}, \qquad \ell_a(\xi) = \xi(a).$$

Thus elements of the Lie algebra can also be regarded as linear observables. The rule

$$\{\ell_a, \ell_b\} = \ell_{[a,b]}$$

transfers the Lie bracket to these observables. Extending it by bilinearity and the Leibniz rule gives a Poisson bracket on

$$\mathcal{O}(\mathfrak{g}^*) = S(\mathfrak{g}).$$

The multiplication in this algebra is still commutative; the bracket is an additional operation.

More explicitly, write $e^1, \dots, e^d$ for the dual basis and define $x_i = \ell_{e_i}$. At $\xi = \sum_k y_k e^k$ one has $x_i(\xi) = y_i$, and

$$\{x_i, x_j\} = \sum_k c_{ij}^k x_k.$$

For arbitrary polynomial observables,

$$\{f, g\}(\xi) = \sum_{i,j,k=1}^{d} c_{ij}^k y_k \frac{\partial f}{\partial x_i}(\xi) \frac{\partial g}{\partial x_j}(\xi). \tag{3.1}$$

Skew-symmetry and Jacobi come from the corresponding Lie algebra identities. In particular, the Jacobiator of a skew biderivation is a derivation in each argument, so checking Jacobi on the generators suffices. This is the *linear Lie-Poisson*, or *Kirillov-Kostant*, structure.

For $\mathfrak{g} = \mathfrak{gl}_N$, the matrix units provide a particularly explicit example. We now work directly with its coordinate formulas, without requiring the abstract construction at every step.

3.2 The bracket on matrix entries

Let

$$A_N = \mathcal{O}_N = \mathbb{C}[x_{ij} : 1 \le i, j \le N], \qquad X = (x_{ij})_{i,j=1}^{N}.$$

Here X is the *universal matrix*: its entries are independent coordinate functions. Evaluating these functions at a numerical matrix recovers that matrix. The algebra A_N is commutative, associative, and unital. Its Poisson bracket is specified on generators by

$$\boxed{\{x_{ij}, x_{kl}\} = \delta_{jk} x_{il} - \delta_{il} x_{kj}.} \tag{3.2}$$

Throughout this talk we keep this sign convention and the earlier convention $D_H f = \{H, f\}$ for Hamiltonian evolution.

Recall that the matrix unit E_{ij} has entries $(E_{ij})_{kl} = \delta_{ik}\delta_{jl}$. Multiplication gives

$$E_{ij}E_{kl} = \delta_{jk}E_{il}, \qquad [E_{ij}, E_{kl}] = \delta_{jk}E_{il} - \delta_{il}E_{kj}.$$

Consequently (3.2) is exactly the Lie-Poisson rule with $x_{ij} = \ell_{E_{ij}}$. It therefore satisfies Jacobi. The unique extension to arbitrary polynomials is

$$\{f, g\} = \sum_{i,j,k,l=1}^{N} \frac{\partial f}{\partial x_{ij}} \frac{\partial g}{\partial x_{kl}} \big(\delta_{jk} x_{il} - \delta_{il} x_{kj}\big). \tag{3.3}$$

In particular, $\{1, f\} = 0$. As in the first talk, bilinearity means that the bracket can be written as a linear map $A_N \otimes_{\mathbb{C}} A_N \to A_N$, sending $f \otimes g$ to $\{f, g\}$.

A first central observable. A polynomial C is called a *Casimir* or *Poisson-central element* if $\{C, f\} = 0$ for all $f \in A_N$. For instance,

$$\{\operatorname{tr} X, x_{kl}\} = \sum_i \big(\delta_{ik} x_{il} - \delta_{il} x_{ki}\big) = x_{kl} - x_{kl} = 0.$$

It is enough to check against coordinate generators, since their products span A_N and the bracket is a derivation.

Example: a 2×2 matrix. Writing $X = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, the six brackets between distinct generators are

$$\begin{aligned} &\{a, b\} = b, && \{a, c\} = -c, && \{a, d\} = 0, \\ &\{b, c\} = a - d, && \{b, d\} = b, && \{c, d\} = -c. \end{aligned}$$

For a nonlinear example, the Leibniz rule gives

$$\{a, b^2 c\} = 2b\{a, b\}c + b^2\{a, c\} = b^2 c.$$

Both $s = a + d$ and $\Delta = ad - bc$ are central. For example,

$$\{\Delta, b\} = a\{d, b\} + \{a, b\}d - b\{c, b\} = -ab + bd - b(d - a) = 0.$$

The general reason for these two Casimirs will emerge below.

3.3 Hamiltonian flows are conjugation flows

For a scalar observable h, the notation $\{h, X\}$ means the matrix whose (k,l) entry is $\{h, x_{kl}\}$. Expanding $X = \sum_{k,l} x_{kl} E_{kl}$ gives

$$\begin{aligned}\{x_{ij}, X\} &= \sum_{k,l} (\delta_{jk} x_{il} - \delta_{il} x_{kj}) E_{kl} \\ &= \sum_{l} x_{il} E_{jl} - \sum_{k} x_{kj} E_{ki} = [E_{ji}, X].\end{aligned}$$

Thus

$$\boxed{\{x_{ij}, X\} = [E_{ji}, X].} \tag{3.4}$$

The transpose of the indices is essential. The first term is row i of X placed in row j, precisely as in the product $E_{ji}X$.

For $H = x_{ij}$, Hamilton's equation and its solution are

$$\dot{X} = [E_{ji}, X], \qquad X(t) = e^{tE_{ji}} X(0) e^{-tE_{ji}}. \tag{3.5}$$

Differentiating the right-hand side checks the equation and the initial condition. For a polynomial f, the scalar chain rule reads

$$\{x_{ij}, f\}(X) = \sum_{k,l} \frac{\partial f}{\partial x_{kl}}(X) [E_{ji}, X]_{kl} = \left.\frac{d}{dt}\right|_{t=0} f(e^{tE_{ji}} X e^{-tE_{ji}}).$$

Thus a scalar observable evolves by the directional derivative along the conjugation flow, while the coordinate matrix evolves by a commutator.

For example, the Hamiltonian $H = a = x_{11}$ in the 2×2 case gives

$$\dot{a} = \dot{d} = 0, \qquad \dot{b} = b, \qquad \dot{c} = -c, \qquad X(t) = \begin{pmatrix} a(0) & e^{t} b(0) \\ e^{-t} c(0) & d(0) \end{pmatrix}.$$

Its trace and determinant are visibly preserved. A Casimir is preserved by *every* Hamiltonian, and its own Hamiltonian vector field is zero.

General Hamiltonians and the sign convention. Define the trace gradient by

$$df_X(V) = \operatorname{tr}\big(\nabla f(X) V\big), \qquad \nabla f(X) = \sum_{i,j} \frac{\partial f}{\partial x_{ij}}(X) E_{ji}.$$

Then the same calculation gives

$$\{H, X\} = [\nabla H(X), X], \qquad \{f, g\} = \operatorname{tr}\big(X[\nabla g, \nabla f]\big). \tag{3.6}$$

To see the second identity, compute $dg_X([\nabla f, X]) = \operatorname{tr}(\nabla g [\nabla f, X])$ and cyclically permute the factors under the trace.

There is a useful convention check here. If $\xi \in \mathfrak{gl}_N^*$ is represented by a matrix Y using $\xi(B) = \operatorname{tr}(YB)$, then $x_{ij}(\xi) = \xi(E_{ij}) = Y_{ji}$, so our coordinate matrix is $X = Y^{\mathsf{T}}$. This explains why formulas written directly for Y with the trace pairing may have the opposite sign. Equation (3.2) fixes our convention unambiguously; the matching trace and tensor formulas appear in [5, Section 2.1.2].

3.4 Powers, traces, and the characteristic polynomial

The scalar Poisson bracket acts entrywise on matrix products and satisfies the matrix-valued Leibniz rule. Therefore

$$\begin{aligned}\{x_{ij}, X^2\} &= [E_{ji}, X]X + X[E_{ji}, X] \\ &= E_{ji}X^2 - X^2E_{ji} = [E_{ji}, X^2].\end{aligned}$$

The middle terms cancel. By induction, for every $r \geq 1$,

$$\{x_{ij}, X^r\} = [E_{ji}, X^r], \qquad \{x_{ij}, \operatorname{tr}(X^r)\} = \operatorname{tr}[E_{ji}, X^r] = 0. \tag{3.7}$$

Thus all power traces $p_r = \operatorname{tr}(X^r)$ are central.

Let v be a scalar indeterminate with zero Poisson bracket with every observable. Define the coefficients c_k by

$$D_X(v) = \det(vI_N + X) = \sum_{k=0}^{N} c_k(X) v^{N-k}, \qquad c_0 = 1. \tag{3.8}$$

With this plus-sign convention, the usual characteristic polynomial is

$$\det(\lambda I_N - X) = \sum_{k=0}^{N} (-1)^k c_k \lambda^{N-k}.$$

In particular, $c_1 = \operatorname{tr} X$ and $c_N = \det X$.

The logarithmic derivative of the determinant. Jacobi's formula is [1]

$$d \det B = \operatorname{tr}(\operatorname{adj}(B)\, dB).$$

For invertible B this becomes

$$d \det B = \det B \, \operatorname{tr}(B^{-1} dB).$$

Applying this formula to $B = vI_N + X$ gives

$$\boxed{\frac{D_X'(v)}{D_X(v)} = \frac{d}{dv} \log D_X(v) = \operatorname{tr}\big((vI_N + X)^{-1}\big).} \tag{3.9}$$

Without the logarithm, the right-hand side must be multiplied by $D_X(v)$. The quotient identity holds wherever $D_X(v) \neq 0$, or as an identity of rational functions; it does not require choosing a global logarithm.

Expanding formally at $v = \infty$ gives

$$\begin{aligned}(vI_N + X)^{-1} &= \sum_{r \geq 0} (-1)^r X^r v^{-r-1}, \\ \frac{D_X'(v)}{D_X(v)} &= \frac{N}{v} - \frac{p_1}{v^2} + \frac{p_2}{v^3} - \frac{p_3}{v^4} + \cdots .\end{aligned}$$

The leading coefficient is $\operatorname{tr} I_N = N$. Multiplying by $D_X(v)$ and comparing coefficients yields Newton's identities:

$$k c_k = \sum_{r=1}^{k} (-1)^{r-1} c_{k-r} p_r \qquad (1 \leq k \leq N). \tag{3.10}$$

For example,

$$c_1 = p_1, \qquad c_2 = \frac{p_1^2 - p_2}{2}, \qquad c_3 = \frac{p_1^3 - 3p_1p_2 + 2p_3}{6}.$$

Each c_k is a polynomial in central elements, hence is central. Equivalently,

$$\{x_{ij}, \det(vI_N + X)\} = 0.$$

Conjugation invariance of the determinant gives another proof using (3.5). The infinitely many traces supply only N algebraically independent generators: Newton's identities and the Cayley-Hamilton identity express the higher traces in terms of the first N.

3.5 Do these functions give the whole center?

For the polynomial Poisson algebra A_N, these Casimirs generate the entire Poisson center:

$$\boxed{Z_{\mathrm{Pois}}(A_N) = \mathbb{C}[c_1, \ldots, c_N] = \mathbb{C}[p_1, \ldots, p_N].} \tag{3.11}$$

We have already proved that the right-hand side is contained in the center. Here is why there are no other polynomial generators.

If f is central, then $\{x_{ij}, f\} = 0$ for every i, j. The flow calculation says that f is constant under conjugation by $e^{tE_{ji}}$. These elementary unipotent and diagonal one-parameter subgroups generate $\mathrm{GL}_N(\mathbb{C})$, so f is conjugation-invariant. Conversely, differentiating conjugation invariance shows that all these brackets vanish. Thus

$$Z_{\mathrm{Pois}}(A_N) = A_N^{\mathrm{GL}_N(\mathbb{C})}.$$

Restrict an invariant polynomial to diagonal matrices $\mathrm{diag}(\lambda_1, \ldots, \lambda_N)$. Permutation matrices make this restriction symmetric in the λ_i, hence a polynomial Q in their elementary symmetric functions. These are exactly the restrictions of $c_1, \ldots, c_N$. Therefore $f = Q(c_1, \ldots, c_N)$ on diagonal matrices, and by invariance on every diagonalizable matrix. Matrices with distinct eigenvalues form a dense open subset, so the two polynomials agree everywhere. Algebraic independence follows by restriction to the algebraically independent elementary symmetric functions. This is the invariant-theory argument in [2, Theorem 1.1]; the second equality in (3.11) follows from (3.10) in characteristic zero.

The word *polynomial* matters. For a larger algebra of smooth or holomorphic observables, functions of the Casimirs need not be polynomials. Also, the characteristic polynomial does not distinguish every conjugacy orbit: different Jordan forms can have identical characteristic polynomials. This does not contradict the description of the polynomial center.

3.6 Symplectic leaves and the number of integrals

The coordinate Hamiltonian vector fields span, at a matrix X, the space

$$\{[B, X] : B \in \mathrm{Mat}_N(\mathbb{C})\}.$$

This is exactly the tangent space to its conjugacy orbit. It follows that the symplectic leaves of this Poisson structure are the conjugacy orbits, equivalently the coadjoint orbits after identifying the dual appropriately. Here a *symplectic leaf* is a maximal connected integral manifold of the distribution spanned by Hamiltonian vector fields. It carries the induced nondegenerate Poisson structure, equivalently a symplectic form. The full space of matrices generally has a degenerate bracket.

Let

$$\mathfrak{z}(X) = \{B \in \mathrm{Mat}_N(\mathbb{C}) : [B, X] = 0\}$$

be the matrix centralizer. The linear map $B \mapsto [B, X]$ has this kernel, so

$$\dim \mathcal{L}_X = N^2 - \dim \mathfrak{z}(X),$$

where $\mathcal{L}_X$ is the conjugacy orbit through X. A matrix is called *regular* if its centralizer has the minimum dimension N. For example, for a diagonal matrix with distinct eigenvalues the centralizer consists of diagonal matrices. A regular leaf therefore has dimension

$$N^2 - N = N(N-1).$$

All dimensions here are complex dimensions. In the algebraic analogue of Hamiltonian mechanics this gives $N(N-1)/2$ degrees of freedom on a regular leaf, and Liouville integrability calls for that many independent commuting Hamiltonians on the leaf.

At regular points the N independent Casimirs provide local transverse coordinates. Their restrictions to any fixed leaf are constant, so they do not provide its nontrivial Hamiltonians. Their numerical values need not distinguish different singular orbits globally. The target number of commuting independent polynomials on the ambient matrix space is

$$N + \frac{N(N-1)}{2} = \frac{N(N+1)}{2}.$$

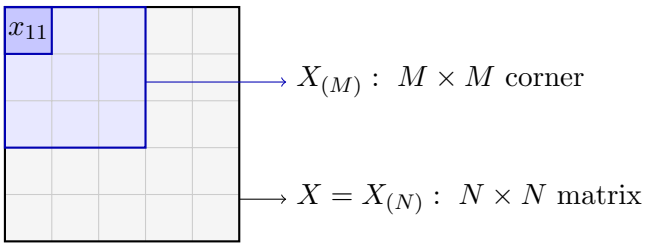
To reach this number, we use invariants of smaller matrix blocks.

3.7 Nested corners and the Gelfand-Tsetlin algebra

For $1 \le M \le N$, let $X_{(M)} = (x_{ij})_{1 \le i,j \le M}$ be the upper-left principal $M \times M$ submatrix. The corresponding coordinate algebras form a chain of Poisson subalgebras:

$$A_1 \subset A_2 \subset \cdots \subset A_N.$$

Indeed, the bracket of entries with indices at most M again involves only such entries.



For each level define

$$\det(vI_M + X_{(M)}) = \sum_{k=0}^{M} v^{M-k} G_{kM}, \qquad G_{0M} = 1. \tag{3.12}$$

Here the second index records the size of the corner and the first records the degree of the coefficient. The first two levels are

$$G_{11} = x_{11}, \qquad G_{12} = x_{11} + x_{22}, \qquad G_{22} = x_{11}x_{22} - x_{12}x_{21}.$$

At level N we recover $G_{kN} = c_k$.

Each G_{kM} belongs to $Z_{\mathrm{Pois}}(A_M)$, but for $M < N$ it need not be central in A_N. For example, $G_{11} = x_{11}$ is central in A_1 while $\{x_{11}, x_{12}\} = x_{12}$ in A_N when $N \geq 2$. Centrality in A_M gives $\{x_{ij}, \det(vI_M + X_{(M)})\} = 0$ for $i, j \leq M$; there is no such assertion for arbitrary indices in the larger algebra.

Nevertheless, coefficients from *all* levels commute with one another. If $M \leq L$, then $G_{kM} \in A_L$ and G_{jL} is central in A_L. Hence

$$\boxed{\{G_{kM}, G_{jL}\} = 0 \quad \text{for all } 1 \leq k \leq M \leq N, \quad 1 \leq j \leq L \leq N.} \tag{3.13}$$

The other ordering follows by skew-symmetry. The algebra

$$\Gamma_N = \mathbb{C}[G_{kM} : 1 \leq k \leq M \leq N]$$

is the classical *Gelfand-Tsetlin algebra*, also called the Gelfand-Zeitlin algebra. It is Poisson-commutative, meaning its elements have zero brackets with one another. This is stronger than ordinary commutativity of multiplication, which already holds throughout A_N.

3.8 Why the coefficients are algebraically independent

There are $1 + 2 + \cdots + N = N(N+1)/2$ nonconstant coefficients. Counting them does not prove independence. In fact, they are *jointly* algebraically independent, not just independent separately at each fixed level. This is part of the Gelfand-Zeitlin theory of Kostant and Wallach [3, Theorem 2.3 and Remark 2.4]. The following differential calculation gives a direct proof of the independence needed here.

It is convenient to replace the characteristic coefficients by the normalized power traces

$$F_{kM}(X) = \frac{1}{k} \operatorname{tr}(X_{(M)}^k), \qquad 1 \leq k \leq M \leq N.$$

At each level, Newton's identities give an invertible triangular polynomial change between the two sets of generators

$$(G_{1M}, \ldots, G_{MM}) \quad \text{and} \quad (F_{1M}, \ldots, F_{MM}).$$

We may therefore prove independence for the F_{kM}.

Evaluate their differentials at the lower shift matrix

$$S = \sum_{i=1}^{N-1} E_{i+1,i} = \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 1 & 0 & \cdots & 0 \\ 0 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \end{pmatrix}.$$

For a variation $V = (v_{ij})$, cyclicity of the trace gives

$$\begin{aligned} (dF_{kM})_S(V) &= \frac{1}{k} \sum_{r=0}^{k-1} \operatorname{tr}\big(S_{(M)}^r V_{(M)} S_{(M)}^{k-1-r}\big) \\ &= \operatorname{tr}\big(S_{(M)}^{k-1} V_{(M)}\big) = \sum_{i=1}^{M-k+1} v_{i,i+k-1}. \end{aligned}$$

Equivalently,

$$(dF_{kM})_S = \sum_{i=1}^{M-k+1} dx_{i,i+k-1}.$$

For $M = k$ this is dx_{1k}; for $M > k$ subtracting the preceding level isolates one coordinate:

$$(dF_{kM})_S - (dF_{k,M-1})_S = dx_{M-k+1,M}.$$

As (k, M) ranges over the triangle, these are precisely all the $N(N+1)/2$ upper-triangular coordinate differentials. They are linearly independent. The subtraction operations are invertible, so the original dF_{kM} are independent as well.

Thus the polynomial map with coordinates F_{kM} has full rank at S. Its image contains a complex analytic open set near its value at S, on which no nonzero polynomial can vanish identically. There can be no polynomial relation among the F_{kM}, and therefore none among the G_{kM}. This proves

$$\Gamma_N \cong \mathbb{C}[t_1, \ldots, t_{N(N+1)/2}].$$

Independence on a leaf. The set where all dG_{kM} are independent is a nonempty Zariski open set, called the *strongly regular locus*. Independence need not hold at every matrix; for example, when $N \geq 2$, the differentials of the coefficients of degree at least two vanish at $X = 0$. Strong regularity also implies regularity: the top-level trace gradients $I, X, \ldots, X^{N-1}$ are independent, so the minimal polynomial has degree N and the centralizer has dimension N.

At a strongly regular point the N independent top-level Casimir differentials span the annihilator of the tangent space to the regular leaf. After restriction to that leaf, the remaining functions

$$G_{kM}, \qquad 1 \leq k \leq M < N,$$

still have independent differentials. Indeed, a relation among their restrictions would express a linear combination of their differentials as a combination of the top-level differentials, contradicting full independence. They supply exactly

$$\sum_{M=1}^{N-1} M = \frac{N(N-1)}{2}$$

commuting Hamiltonians, half the leaf dimension. This is the Gelfand-Tsetlin integrable system on the strongly regular part of a regular leaf; see also [4, Section 2].

For $N = 2$, the family is simply (a, s, Δ). If $s^2 - 4\Delta \neq 0$, fixing the two Casimirs gives a two-dimensional complex conjugacy orbit; a supplies its one Hamiltonian. Its explicit flow was computed in Section 3.3. At a diagonal matrix its vector field vanishes, which illustrates why differential independence is a generic condition rather than a condition at every point.

3.9 Tensor notation: what the bracket of two matrices means

Let $V = \mathbb{C}^N$. To organize all N^4 scalar brackets at once, work in

$$\mathrm{End}(V) \otimes \mathrm{End}(V) \otimes A_N.$$

The first two factors record matrix positions; the third contains the polynomial coefficients. For numerical matrices A, B, write

$$A_1 = A \otimes I_N, \qquad B_2 = I_N \otimes B, \qquad A_1 B_2 = A \otimes B.$$

The alternative notation $\overset{1}{A}, \overset{2}{B}$ uses superscripts as tensor-factor labels, not powers. In particular,

$$X_1 = \sum_{i,j} (E_{ij} \otimes I_N) x_{ij}, \qquad X_2 = \sum_{k,l} (I_N \otimes E_{kl}) x_{kl}.$$

They are two placements of the *same* coordinate matrix, not matrices with independent sets of entries.

For matrices $A = (a_{ij})$ and $B = (b_{kl})$ whose entries are observables, define the *tensor Poisson bracket* by

$$\boxed{\left\{A_1 \overset{\otimes}{,} B_2\right\} := \sum_{i,j,k,l} (E_{ij} \otimes E_{kl}) \{a_{ij}, b_{kl}\}.} \tag{3.14}$$

The symbol $\otimes$ above the comma is Tarasov's notation: bracket the scalar coefficients and retain both auxiliary matrix slots. The labels $1, 2$ specify their placements. In particular, $\left\{X_1 \overset{\otimes}{,} X_2\right\}$ uses the same coordinate matrix in both slots. This is an entrywise Poisson operation; the commutator $[A_1, B_2]$ is a separate matrix operation.

Exchanging the two matrix arguments also exchanges their tensor slots. Scalar antisymmetry therefore gives

$$\left\{A_1 \overset{\otimes}{,} B_2\right\} = -P \left\{B_1 \overset{\otimes}{,} A_2\right\} P,$$

where P is the permutation operator defined below. All brackets with two matrix arguments use the tensor comma from this point on. Brackets between scalar observables keep an ordinary comma, as do scalar-to-matrix brackets such as $\{h, X\}$, which are evaluated entrywise and have only one matrix slot.

In the ordered tensor basis $e_i \otimes e_j$, one has

$$\left(\left\{X_1 \overset{\otimes}{,} X_2\right\}\right)_{(i,j),(k,l)} = \{x_{ik}, x_{jl}\}. \tag{3.15}$$

The row index is the pair (i, j) and the column index is (k, l).

There are two uses of tensor products here. The map $A_N \otimes A_N \to A_N$ encodes *bilinearity of the scalar bracket*. The factors $\mathrm{End}(V) \otimes \mathrm{End}(V)$ in (3.14) keep track of *two matrix indices*. These roles should be distinguished. In fact, ordinary matrix multiplication gives $X_1 X_2 = X_2 X_1$, whereas their tensor Poisson bracket is generally nonzero.

The permutation operator. Let $P = P^{12} : V \otimes V \to V \otimes V$ be the flip,

$$P(v \otimes w) = w \otimes v, \qquad P = \sum_{i,j=1}^{N} E_{ij} \otimes E_{ji}. \tag{3.16}$$

Indeed, applying the sum to $e_a \otimes e_b$ leaves only $i = b, j = a$, producing $e_b \otimes e_a$. For $N = 2$, in the basis $(e_1 \otimes e_1, e_1 \otimes e_2, e_2 \otimes e_1, e_2 \otimes e_2)$,

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Applying these operators to simple tensors proves

$$P^2 = I, \qquad PA_1 = A_2 P, \qquad PA_2 = A_1 P.$$

Consequently,

$$[P, X_1 + X_2] = 0, \qquad [P, X_1 X_2] = 0. \tag{3.17}$$

For the second identity use $PX_1X_2 = X_2PX_2 = X_2X_1P = X_1X_2P$.

Packing the matrix-entry bracket into one equation. Using (3.4) in the second tensor slot,

$$\begin{aligned}\{X_1 \overset{\otimes}{,} X_2\} &= \sum_{i,j}(E_{ij}\otimes I_N)\{x_{ij}, X_2\}\\ &= \sum_{i,j}(E_{ij}\otimes I_N)[I_N\otimes E_{ji}, X_2]\\ &= \left[\sum_{i,j} E_{ij}\otimes E_{ji}, X_2\right].\end{aligned}$$

In the last line, $E_{ij}\otimes I_N$ commutes with X_2. Thus

$$\boxed{\{X_1 \overset{\otimes}{,} X_2\} = [P, X_2] = [X_1, P].} \tag{3.18}$$

The second equality is (3.17). As an entrywise check,

$$[P, X_2]_{(i,k),(j,l)} = \delta_{kj}x_{il} - \delta_{il}x_{kj} = \{x_{ij}, x_{kl}\}.$$

So no information was lost: (3.18) and (3.2) are equivalent descriptions of the same bracket.

3.10 The two spectral-parameter exercises

Let u, v be scalar parameters with zero Poisson brackets with all observables. We may work over the rational function field $\mathbb{C}(u, v)$; for numerical parameters assume $u, v \neq 0$ where division by them occurs, and $u \neq v$ in both displayed commutator formulas. Write

$$r(u,v) = \frac{P}{u-v}, \qquad B = [P, X_2] = -[P, X_1].$$

The parameters are not dynamical variables. Scalar terms such as u in matrix expressions mean u times the appropriate identity.

Exercise (a): the linear form. Show that (3.18) is equivalent to

$$\boxed{\left\{\frac{X_1}{u} \overset{\otimes}{,} \frac{X_2}{v}\right\} = \left[\frac{P}{u-v}, \frac{X_1}{u} + \frac{X_2}{v}\right].} \tag{3.19}$$

Exercise (b): the quadratic form. Show that (3.18) is also equivalent to

$$\boxed{\{uI + X_1 \overset{\otimes}{,} vI + X_2\} = \left[\frac{P}{u-v}, (uI + X_1)(vI + X_2)\right].} \tag{3.20}$$

Their solutions and interpretation as linear and quadratic r-matrix relations are given in Section 4.1. The Gelfand-Tsetlin construction above will reappear there as a limit of a new family of commuting subalgebras.

4 Shift of Argument Subalgebra

Speaker: Vitaly Tarasov (IU Indianapolis)
Date: October 9, 2026
Tarasov's second talk; seminar Talk 4.

The previous lecture constructed commuting functions from nested matrix corners. We now obtain another family from one matrix depending on a spectral parameter. A constant matrix C breaks conjugation symmetry: the resulting spectral invariants still commute with each other, but need not be Casimirs. A degeneration of C recovers the corner construction. For invertible C, a determinant identity identifies the new family with invariant polynomials evaluated at the shifted argument $X + tC^{-1}$.

4.1 From the previous exercises to Yang's rational matrix

We keep exactly the sign convention of the first Tarasov lecture:

$$\{x_{ij}, x_{kl}\} = \delta_{jk}x_{il} - \delta_{il}x_{kj}, \qquad B := \{X_1 \overset{\otimes}{,} X_2\} = [P, X_2] = -[P, X_1].$$

Here $X_1 = X \otimes I$, $X_2 = I \otimes X$, and $P = \sum E_{ij} \otimes E_{ji}$ exchanges the two auxiliary vector spaces. All scalar parameters below have zero bracket with all observables. We use Tarasov's tensor comma for matrix Poisson brackets throughout. Its entrywise definition is (3.14); scalar brackets still use an ordinary comma.

The rational function

$$r_{12}(u, v) = \frac{P}{u - v}$$

is called *Yang's classical rational r-matrix*. It organizes the matrix brackets into a commutator. The word classical refers to the Poisson algebra of the entries: the auxiliary matrix commutator is not a quantum commutator of observables.

Solution of the linear exercise. Since $[P, X_1] = -B$ and $[P, X_2] = B$,

$$\left[\frac{P}{u - v}, \frac{X_1}{u} + \frac{X_2}{v}\right] = \frac{-B/u + B/v}{u - v} = \frac{B}{uv} = \left\{\frac{X_1}{u} \overset{\otimes}{,} \frac{X_2}{v}\right\}.$$

Multiplication by uv gives the converse. Thus $L(u) = X/u$ satisfies

$$\{L_1(u) \overset{\otimes}{,} L_2(v)\} = [r_{12}(u, v), L_1(u) + L_2(v)]. \tag{4.1}$$

Solution of the quadratic exercise. Put $T_0(u) = uI + X$. The scalar terms do not contribute to the bracket. Also $[P, X_1X_2] = 0$, so

$$\begin{aligned}[P, (uI + X_1)(vI + X_2)] &= v[P, X_1] + u[P, X_2] + [P, X_1X_2] \\ &= (u - v)B.\end{aligned}$$

After division by $u - v$, this is precisely

$$\{T_{0,1}(u) \overset{\otimes}{,} T_{0,2}(v)\} = [r_{12}(u, v), T_{0,1}(u)T_{0,2}(v)]. \tag{4.2}$$

The same computation proves equivalence with the original entry bracket. These are identities of rational functions; numerical evaluation requires avoiding poles. A denominator cleared identity continues to make sense where the original rational expression is undefined.

4.2 In what sense is the linear relation a limit?

To recover the linear relation from the quadratic one, rescale the spectral parameters by $u \mapsto u/\varepsilon$ and $v \mapsto v/\varepsilon$. Normalize the matrix and track the order in ε. Define

$$S_\varepsilon(u) = \frac{\varepsilon}{u} T_0(u/\varepsilon) = I + \varepsilon L(u), \qquad L(u) = \frac{X}{u}.$$

Since $r(u/\varepsilon, v/\varepsilon) = \varepsilon r(u,v)$, the quadratic equation becomes

$$\big\{ S_{\varepsilon,1}(u) \mathbin{\overset{\otimes}{,}} S_{\varepsilon,2}(v) \big\} = \varepsilon [r(u,v), S_{\varepsilon,1}(u) S_{\varepsilon,2}(v)].$$

The left side is $\varepsilon^2 \big\{ L_1 \mathbin{\overset{\otimes}{,}} L_2 \big\}$; expanding the right side gives

$$\varepsilon^2 [r, L_1 + L_2] + \varepsilon^3 [r, L_1 L_2].$$

Divide by ε^2 before taking the limit. The result is (4.1). In this particular one-matrix example the last commutator vanishes identically, because $L_1 L_2 = X_1 X_2/(uv)$. The limit illustrates the general linearization mechanism even though both formulas here encode the same linear bracket. Simply setting $\varepsilon = 0$ without the division would give only $0 = 0$.

4.3 Why spectral traces Poisson-commute

Now let $T(u)$ be a matrix with entries rational in u, with coefficients in a commutative Poisson algebra and fixed scalar poles, satisfying the quadratic relation

$$\big\{ T_1(u) \mathbin{\overset{\otimes}{,}} T_2(v) \big\} = [r_{12}(u,v), T_1(u) T_2(v)]. \tag{4.3}$$

The matrix r is numerical (its entries do not depend on the phase-space variables). We prove that

$$\boxed{\{ \operatorname{tr} T(u)^k, \operatorname{tr} T(v)^m \} = 0 \quad (k, m \geq 1).} \tag{4.4}$$

Here $\operatorname{tr}_{12} = \operatorname{tr} \otimes \operatorname{tr}$ traces both auxiliary spaces, so $\operatorname{tr}_{12}(A_1 B_2) = \operatorname{tr} A \, \operatorname{tr} B$.

Proof with all Leibniz terms retained. Set $U = T_1(u)$, $V = T_2(v)$, and $Q = \big\{ U \mathbin{\overset{\otimes}{,}} V \big\}$. Since they occupy different matrix slots and their coefficients commute, $UV = VU$. The Leibniz rule, however, does not say that Q commutes with U or V. The correct expansion is

$$\Big\{ U^k \mathbin{\overset{\otimes}{,}} V^m \Big\} = \sum_{i=0}^{k-1} \sum_{j=0}^{m-1} U^i V^j Q V^{m-1-j} U^{k-1-i}.$$

After taking the trace, cyclicity makes all km terms equal:

$$\begin{aligned} \{ \operatorname{tr} T(u)^k, \operatorname{tr} T(v)^m \} &= km \, \operatorname{tr}_{12} \big(U^{k-1} V^{m-1} [r, UV] \big) \\ &= km \, \operatorname{tr}_{12} \big(r U^k V^m - U^k V^m r \big) = 0. \end{aligned}$$

This uses only cyclicity, not diagonalizability of either matrix.

Taking traces is essential. Before taking traces, one cannot generally replace $\big\{ U \mathbin{\overset{\otimes}{,}} V^2 \big\}$ by $[r, UV^2]$. Indeed,

$$\big\{ U \mathbin{\overset{\otimes}{,}} V^2 \big\} = QV + VQ = [r, UV^2] + V[r, U]V.$$

The extra term need not vanish. Likewise the untraced identity $\Big\{ U^k \mathbin{\overset{\otimes}{,}} V^m \Big\} = [r, U^k V^m]$ is not valid in general. The theorem about traces is nevertheless correct, by the calculation above. Cyclicity removes the additional terms only after taking traces.

4.4 From traces to determinants and exterior powers

For a numerical or polynomial matrix A, let $\bigwedge^k A$ denote the induced map on $\bigwedge^k \mathbb{C}^N$:

$$(\bigwedge^k A)(v_1 \wedge \cdots \wedge v_k) = Av_1 \wedge \cdots \wedge Av_k.$$

It is an exterior power, not the ordinary matrix power A^k. In the basis $e_{i_1} \wedge \cdots \wedge e_{i_k}$ its diagonal entries are the principal $k \times k$ minors of A. Hence

$$e_k(A) := \operatorname{tr}(\bigwedge^k A) = \sum_{|I|=k} \det A_{I,I}, \qquad \det(tI + A) = \sum_{k=0}^{N} t^{N-k} e_k(A). \tag{4.5}$$

We set $e_0 = 1$. For example,

$$e_1(A) = \sum_i a_{ii}, \qquad e_2(A) = \sum_{i<j} (a_{ii}a_{jj} - a_{ij}a_{ji}) = \frac{(\operatorname{tr} A)^2 - \operatorname{tr}(A^2)}{2}.$$

The determinant identity follows directly by expanding in principal minors, or by checking the elementary symmetric functions of eigenvalues on the dense set of diagonalizable matrices.

To pass from commuting traces to commuting determinants, put $p_j(u) = \operatorname{tr} T(u)^j$ and $e_k(u) = e_k(T(u))$. Newton's identities give

$$k e_k(u) = \sum_{j=1}^{k} (-1)^{j-1} e_{k-j}(u) p_j(u).$$

Thus every $e_k(u)$ is a polynomial in $p_1(u), \ldots, p_k(u)$. Since all $p_j(u)$ commute with all $p_l(v)$, the Leibniz rule implies $\{e_k(u), e_l(v)\} = 0$. Consequently

$$\boxed{\{\det(tI + T(u)), \det(sI + T(v))\} = 0.} \tag{4.6}$$

The scalar parameters t, s, u, v are independent and have zero Poisson brackets. Taking Laurent coefficients in u, v, or clearing their fixed scalar denominators and comparing polynomial coefficients, gives a family of pairwise commuting scalar observables. Each member is conserved under a Hamiltonian chosen from this commuting family.

4.5 A constant twist supplies noncentral Hamiltonians

The choice $T_0(u) = uI + X$ alone gives the familiar Casimirs. To get more functions, fix *any* constant matrix $C \in \operatorname{Mat}_N(\mathbb{C})$ and put

$$T_C(u) = C(uI + X).$$

No invertibility assumption is needed here. Since $P(C \otimes C) = (C \otimes C)P$, one has $[r, C_1C_2] = 0$. Also C_2 commutes with $T_{0,1}$ and C_1 with $T_{0,2}$. Therefore

$$\begin{aligned} \{T_{C,1}(u) \overset{\otimes}{,} T_{C,2}(v)\} &= C_1C_2 \{T_{0,1}(u) \overset{\otimes}{,} T_{0,2}(v)\} \\ &= C_1C_2[r, T_{0,1}(u)T_{0,2}(v)] \\ &= [r, T_{C,1}(u)T_{C,2}(v)]. \end{aligned}$$

The trace and determinant theorems apply. Define

$$F_C(t,u) = \det(tI + C(uI + X)), \qquad \mathcal{B}_C = \mathbb{C}[\text{coefficients of } F_C(t,u) \text{ in } t,u]. \tag{4.7}$$

Then $\mathcal{B}_C$ is Poisson-commutative. Equivalently one may use the coefficients of $\operatorname{tr}(T_C(u)^k)$, $1 \le k \le N$, since Newton's identities work in both directions.

For a fixed C, commutativity means that every coefficient commutes with every other coefficient belonging to that same C. There is no claim that families for two arbitrary different twists commute with each other. Nor are these functions generally central in the full algebra A_N. For example, if $C = \operatorname{diag}(c_1, \dots, c_N)$, then

$$\{\operatorname{tr}(CX), x_{ij}\} = (c_i - c_j)x_{ij}.$$

This can generate nontrivial motion while preserving all functions in $\mathcal{B}_C$.

4.6 The two-by-two example

Write

$$X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad C = \begin{pmatrix} c_1 & 0 \\ 0 & c_2 \end{pmatrix}.$$

Direct expansion, including every term in both parameters, gives

$$\begin{aligned} F_C(t,u) = t^2 &+ t\big((c_1 + c_2)u + c_1 a + c_2 d\big) \\ &+ c_1 c_2\big(u^2 + u(a+d) + ad - bc\big). \end{aligned}$$

For $c_1 c_2 \neq 0$ its nonconstant coefficients generate

$$H = c_1 a + c_2 d, \qquad s = a + d, \qquad \Delta = ad - bc.$$

The last two are Casimirs, so all three Poisson-commute. If $c_1 \neq c_2$, then $a = (H - c_2 s)/(c_1 - c_2)$ and $d = s - a$. Thus $\mathcal{B}_C = \mathbb{C}[a, s, \Delta]$, which is the two-by-two Gelfand-Tsetlin algebra. The three differentials are independent, for example at $b = 1, c = 0$: ds, dH span da, dd, while $d\Delta = d\,da + a\,dd - dc$ adds a third direction.

On a generic two-dimensional leaf with s, Δ fixed, H supplies one Hamiltonian. With our convention $\dot{f} = \{H, f\}$,

$$\dot{a} = \dot{d} = 0, \qquad \dot{b} = (c_1 - c_2)b, \qquad \dot{c} = -(c_1 - c_2)c.$$

Hence $b(\tau) = e^{(c_1 - c_2)\tau} b(0)$ and $c(\tau) = e^{-(c_1 - c_2)\tau} c(0)$. The time variable τ is distinct from the determinant parameter t.

If $c_1 = c_2 \neq 0$, H is a multiple of the trace and this family gives only the center. If C is singular, the displayed expansion still holds, but some coefficients disappear; for instance $C = \operatorname{diag}(1, 0)$ gives only $\mathbb{C}[a]$. These cases explain why commutativity alone does not imply a complete integrable system.

4.7 Recovering Gelfand-Tsetlin by a normalized limit

Take

$$C_\varepsilon = \operatorname{diag}(1, \varepsilon, \varepsilon^2, \dots, \varepsilon^{N-1}).$$

Let $Q_{k,\varepsilon}(u)$ be the coefficient of t^{N-k} in $F_{C_\varepsilon}(t,u)$. The principal-minor formula (4.5) gives the exact expression

$$Q_{k,\varepsilon}(u) = \sum_{\substack{I\subset\{1,\dots,N\}\\ |I|=k}} \varepsilon^{\sum_{i\in I}(i-1)} \det\big((uI+X)_{I,I}\big). \tag{4.8}$$

The smallest exponent is $0+1+\cdots+(k-1) = k(k-1)/2$, attained uniquely by $I=\{1,\dots,k\}$. Therefore

$$\boxed{\lim_{\varepsilon\to 0} \varepsilon^{-k(k-1)/2} Q_{k,\varepsilon}(u) = \det(uI_k + X_{(k)}).} \tag{4.9}$$

The normalization is essential: plugging $\varepsilon = 0$ directly into F_C would retain only $t^N + t^{N-1}(u+x_{11})$. Thus the leading terms must be extracted *separately at each power of* t. Each coefficient has its own vanishing order in ε.

For $N=3$, let $D_{ij}(u) = (u+x_{ii})(u+x_{jj}) - x_{ij}x_{ji}$. Then exactly

$$\begin{aligned} Q_{1,\varepsilon}(u) &= (u+x_{11}) + \varepsilon(u+x_{22}) + \varepsilon^2(u+x_{33}),\\ Q_{2,\varepsilon}(u) &= \varepsilon D_{12}(u) + \varepsilon^2 D_{13}(u) + \varepsilon^3 D_{23}(u),\\ Q_{3,\varepsilon}(u) &= \varepsilon^3 \det(uI_3 + X). \end{aligned}$$

Thus the three normalizations are $1, \varepsilon^{-1}, \varepsilon^{-3}$.

All coefficients of the normalized polynomials commute for each nonzero ε. They have polynomial limits, so their brackets also tend to zero. Extracting the coefficients of u in (4.9) recovers precisely the G_{jk} of (3.12). This gives another proof of their commutativity and explains the link between the two talks.

There is also an independence consequence. For each k, the coefficients of $u^0,\dots,u^{k-1}$ in the normalized $Q_{k,\varepsilon}$ tend to the k nonconstant corner invariants. Their joint differentials are independent at the shift matrix S from Section 3.8. A nonzero Jacobian minor at $\varepsilon = 0$ stays nonzero for all sufficiently small nonzero ε. Thus the twisted family has $N(N+1)/2$ independent coefficients for such twists, and hence for generic C. This argument proves generic completeness, without claiming completeness for every C.

4.8 Why this is called the shift-of-argument algebra

Suppose now that C is invertible and set $A = C^{-1}$. Factoring C on the left gives the exact identity

$$\boxed{F_C(t,u) = \det C \, \det(uI + X + tA).} \tag{4.10}$$

Indeed $tI + C(uI+X) = C(tC^{-1} + uI + X)$. No commutation of C and X is used. The fixed matrix A specifies a direction in the vector space of matrices, and $X \mapsto X + tA$ is translation in that direction.

Recall $c_k(X)$ from (3.8). Expand

$$c_k(X+tA) = \sum_{j=0}^{k} t^j f_{k,j}^A(X), \qquad f_{k,j}^A(X) = \frac{1}{j!}\partial_A^j c_k(X), \quad \partial_A = \sum_{a,b} a_{ab}\frac{\partial}{\partial x_{ab}}. \tag{4.11}$$

The algebra

$$\mathcal{A}_A = \mathbb{C}[f_{k,j}^A : 1\le k\le N,\ 0\le j<k]$$

is the *shift-of-argument*, or *Mishchenko-Fomenko*, subalgebra. The omitted term $f_{k,k}^A = c_k(A)$ is constant. Thus there are $\sum_{k=1}^N k = N(N+1)/2$ proposed nonconstant generators. For invertible

A, (4.10) identifies $\mathcal{A}_A = \mathcal{B}_{A^{-1}}$, and proves Poisson commutativity. The brackets are polynomial in the entries of A; since they vanish on the dense open set of invertible A, they vanish for every A.

Equivalently, take all coefficients of all shifted invariant polynomials $f(X+tA)$, $f \in Z_{\mathrm{Pois}}(A_N)$. The basic invariants c_k suffice because the center is generated by them. Using traces instead gives the same algebra by Newton's identities. This is the usual argument-shift construction described in [6].

Commutative does not mean central. The terms with $j = 0$ are the original Casimirs. The additional coefficients usually are not Casimirs. Nor is translation by tA generally a Poisson automorphism of the original Lie-Poisson bracket: for $y_{ij} = x_{ij} + ta_{ij}$,

$$\{y_{ij}, y_{kl}\} = \delta_{jk} y_{il} - \delta_{il} y_{kj} - t(\delta_{jk} a_{il} - \delta_{il} a_{kj}).$$

For nonzero t the last term vanishes for every index precisely when A is scalar. The shift method's content is the mutual commutativity of the shifted invariant coefficients, not preservation of all brackets under translation.

Completeness and exceptional directions. For regular semisimple A, meaning a matrix with distinct eigenvalues, the shift algebra has transcendence degree $N(N+1)/2$. This is the standard completeness result for the argument-shift method; the semisimple statement in [6] applies to $\mathfrak{sl}_N$, and adjoining the central trace direction gives the $\mathfrak{gl}_N$ count. The normalized-limit argument above proves the generic case directly in our coordinates. On the open locus of joint independence, fixing the N Casimirs leaves $N(N-1)/2$ independent commuting Hamiltonians on a regular leaf. For $A = \alpha I$, all shifted invariants are polynomials in the original Casimirs, so the algebra reduces to the center. Counting proposed generators is not an independence proof for an exceptional direction.

4.9 A generating polynomial for the commuting Hamiltonians

The commuting Hamiltonians can be collected in one polynomial. Continue with C invertible and $A = C^{-1}$. For $u \neq 0$, set $z = t/u$ and $w = 1/u$. Then

$$F_C(t,u) = u^N \det C\, \mathcal{G}_A(z,w), \qquad \mathcal{G}_A(z,w) = \det(I + zA + wX). \tag{4.12}$$

Homogeneity of c_k gives the useful coefficient dictionary

$$\mathcal{G}_A(z,w) = \sum_{k=0}^{N} \sum_{j=0}^{k} z^j w^{k-j} f^A_{k,j}(X), \qquad f^A_{0,0} = 1.$$

It is therefore a generating polynomial for the same commuting family. Equivalently, for $A = C^{-1}$ we have

$$\mathcal{G}_A(z, 1/u) = \det(I + zA + X/u).$$

Different powers of z and u^{-1} label individual observables.

For instance, in size two, direct expansion yields

$$\begin{aligned}\det(uI + X + tA) = u^2 + u(\operatorname{tr} X + t \operatorname{tr} A) + \det X \\ + t\big(\operatorname{tr} X \operatorname{tr} A - \operatorname{tr}(XA)\big) + t^2 \det A.\end{aligned}$$

The nonconstant generators may thus be taken as $\operatorname{tr} X$, $\det X$, and $\operatorname{tr}(XA)$, after using the first generator to change the third. This agrees with the earlier twisted example when $A = C^{-1}$.

4.10 The classical Yang-Baxter equation

The classical Yang-Baxter equation supplies a compatibility condition for r-matrix Poisson brackets. Work now on $V^{\otimes 3}$. The subscripts $12, 13, 23$ indicate the two tensor factors on which an operator acts, leaving the remaining one unchanged. For $r_{12}(u,v) = P_{12}/(u-v)$, skew-symmetry reads

$$r_{12}(u,v) = -r_{21}(v,u), \qquad r_{21}(v,u) := P_{12} r_{12}(v,u) P_{12}.$$

It accounts for exchanging the two matrix arguments in the tensor Poisson bracket. The *classical Yang-Baxter equation* is

$$\begin{aligned}[r_{12}(u,v), r_{13}(u,w)] + [r_{12}(u,v), r_{23}(v,w)] \\ + [r_{13}(u,w), r_{23}(v,w)] = 0.\end{aligned} \tag{4.13}$$

Direct verification for the rational matrix. Set $K = [P_{12}, P_{13}]$. Applying the permutations to a simple tensor shows

$$[P_{12}, P_{23}] = -K, \qquad [P_{13}, P_{23}] = K.$$

Substitution into (4.13) gives

$$K\left(\frac{1}{(u-v)(u-w)} - \frac{1}{(u-v)(v-w)} + \frac{1}{(u-w)(v-w)}\right) = 0,$$

since the numerator over the common denominator is $(v-w) - (u-w) + (u-v) = 0$.

For a numerical skew r, this equation is a sufficient compatibility condition for the Jacobi identity of the linear and quadratic r-matrix brackets. In the quadratic case a repeated Leibniz expansion of the Jacobiator reduces, up to the ordering convention for its cyclic sum, to the commutator of the left-hand side of (4.13) with $T_1(u)T_2(v)T_3(w)$. Thus it vanishes. The specific brackets used here already satisfy Jacobi because they were built from copies of the Lie-Poisson algebra; the r-matrix formulation makes this compatibility portable to other examples. For the general framework see [7, Sections 4.6 and 4.8].

4.11 Preview: several sites, products, and sums

Consider M independent copies of the matrix Poisson algebra. Let $X^{(a)} = (x_{ij}^{(a)})$, $1 \le a \le M$, with

$$\{x_{ij}^{(a)}, x_{kl}^{(b)}\} = \delta_{ab}\big(\delta_{jk} x_{il}^{(a)} - \delta_{il} x_{kj}^{(a)}\big). \tag{4.14}$$

Entries from different copies have zero Poisson brackets. The index a is usually called a *site*; it is distinct from the auxiliary matrix indices i, j and from the tensor-slot labels $1, 2$. Choose pairwise distinct fixed parameters $z_1, \dots, z_M$.

Multiplicative construction. Each factor $K^{(a)}(u) = I + X^{(a)}/(u - z_a)$ satisfies the quadratic relation with $r(u,v) = P/(u-v)$: divide the one-site relation by $(u-z_a)(v-z_a)$ and use $(u-z_a) - (v-z_a) = u - v$. If K and J are two such independent factors, the Leibniz rule gives

$$\begin{aligned}\big\{K_1 J_1 \mathbin{\overset{\otimes}{,}} K_2 J_2\big\} &= \big\{K_1 \mathbin{\overset{\otimes}{,}} K_2\big\} J_1 J_2 + K_1 K_2 \big\{J_1 \mathbin{\overset{\otimes}{,}} J_2\big\} \\ &= [r, K_1 K_2] J_1 J_2 + K_1 K_2 [r, J_1 J_2] \\ &= [r, K_1 K_2 J_1 J_2] = [r, (KJ)_1 (KJ)_2].\end{aligned}$$

The middle terms cancel. The rearrangements move only factors in different auxiliary slots; the matrices $K(u)$ and $J(u)$ themselves need not commute. Induction and the constant-twist calculation show that

$$T(u) = C \prod_{a=1}^{M} \left(I + \frac{X^{(a)}}{u - z_a} \right) \tag{4.15}$$

satisfies (4.3). The product is taken in the displayed order $a = 1, \ldots, M$. Its spectral coefficients commute by the theorem.

Additive construction. For any fixed matrix A define

$$L(u) = A + \sum_{a=1}^{M} \frac{X^{(a)}}{u - z_a}. \tag{4.16}$$

Independence removes mixed-site brackets. Applying the linear exercise at each shifted pair of parameters gives

$$\begin{aligned} \big\{ L_1(u) \overset{\otimes}{,} L_2(v) \big\} &= \sum_a \frac{[P, X_2^{(a)}]}{(u - z_a)(v - z_a)} \\ &= [r(u, v), L_1(u) + L_2(v)], \end{aligned}$$

because $[P, A_1 + A_2] = 0$ for every constant A. The same trace argument as before now uses

$$\operatorname{tr}_{12}\big(U^{k-1} V^{m-1} [r, U + V] \big) = 0$$

for $U = L_1(u)$, $V = L_2(v)$: the prefactor commutes with both U and V, and cyclicity cancels each commutator separately. Thus the spectral coefficients of $L(u)$ also form a commuting family. In this multisite preview A and C may be chosen independently; they need not be inverses.

The additive construction is the rational Gaudin model, while the ordered product gives a rational XXX-type spin-chain construction. Their first Laurent coefficients, Gaudin Hamiltonians, and canonical duality are worked out in Section 5. XXZ constructions use a trigonometric structure; see [8, Sections 3.1–3.3].

5 Classical Gaudin Models

Speaker: Vitaly Tarasov (IU Indianapolis)
Date: October 16, 2026
Tarasov's third talk; seminar Talk 5.

Spectral invariants produce commuting observables in two ways: from a product of local matrices, or from a sum of matrices with simple poles. We first relate these constructions by a small-parameter limit. We then expand their characteristic determinants, keeping track of the difference between expansion in the spectral parameter and expansion in the characteristic variable. The residues yield explicit Hamiltonians. A canonical realization by rank-one matrices finally exchanges the size of the matrices with the number of sites.

5.1 Linear and quadratic brackets, and their limiting relation

Write P_{12} for the permutation of the two auxiliary copies of $\mathbb{C}^N$, and put $r_{12}(u,v) = P_{12}/(u-v)$. The two relations are

$$\{T_1(u) \overset{\otimes}{,} T_2(v)\} = [r_{12}(u,v), T_1(u)T_2(v)], \tag{5.1}$$

$$\{L_1(u) \overset{\otimes}{,} L_2(v)\} = [r_{12}(u,v), L_1(u) + L_2(v)]. \tag{5.2}$$

The braces retain both matrix slots, as in (3.14). All entries belong to a commutative Poisson algebra. The commutator acts only in the auxiliary matrix spaces.

Keep the coefficient Poisson algebra fixed and treat ε as a central formal parameter. Suppose a family of quadratic relations has expansions

$$T_\varepsilon(u) = I + \varepsilon L(u) + O(\varepsilon^2), \qquad r_\varepsilon(u,v) = \varepsilon r_{12}(u,v) + O(\varepsilon^2).$$

The first nonzero term of the bracket is $\varepsilon^2 \{L_1 \overset{\otimes}{,} L_2\}$. On the right,

$$[\varepsilon r_{12} + O(\varepsilon^2), I + \varepsilon(L_1 + L_2) + O(\varepsilon^2)] = \varepsilon^2 [r_{12}, L_1 + L_2] + O(\varepsilon^3).$$

Comparing the coefficients of ε^2 gives the linear relation. The scaling of r is part of this limit; it cannot be omitted.

For one matrix X with bracket (3.2), take $C_\varepsilon = I + \varepsilon B$ and normalize

$$\frac{\varepsilon}{u} C_\varepsilon(uI/\varepsilon + X) = (I + \varepsilon B)(I + \varepsilon X/u) = I + \varepsilon(B + X/u) + O(\varepsilon^2).$$

The spectral rescaling gives $r(u/\varepsilon, v/\varepsilon) = \varepsilon r(u,v)$. Thus the constant term B of the linear Lax matrix arises from a twist approaching the identity.

Both brackets imply commutativity of the corresponding characteristic determinants. For the quadratic relation this is (4.6). For the linear relation, put $U = L_1(u)$ and $V = L_2(v)$. The Leibniz rule and cyclicity give

$$\{\operatorname{tr} L(u)^k, \operatorname{tr} L(v)^m\} = km \operatorname{tr}_{12}\big(U^{k-1} V^{m-1} [r, U+V]\big) = 0.$$

Indeed $U^{k-1}V^{m-1}$ commutes with U and V, so cyclicity cancels each commutator separately. Newton's identities then imply

$$\{\det(tI + L(u)), \det(sI + L(v))\} = 0. \tag{5.3}$$

The determinant scaling underlying the limit is, exactly,

$$\det\big((t+1)I+\varepsilon L(u)\big)=\varepsilon^N\det\big(t'I+L(u)\big),\qquad t'=\frac{t+1}{\varepsilon}. \tag{5.4}$$

The factor is ε^N, since a factor ε is removed from each of N rows. To retain arbitrary t' in the limit, set $t=-1+\varepsilon t'$ before letting ε tend to zero.

5.2 Independent sites and the order of the local factors

Let $X^{(a)}=(x_{ij}^{(a)})$, $a=1,\dots,M$, satisfy

$$\{x_{ij}^{(a)},x_{kl}^{(b)}\}=\delta_{ab}(\delta_{jk}x_{il}^{(a)}-\delta_{il}x_{kj}^{(a)}). \tag{5.5}$$

The factor δ_{ab} expresses *ultralocality*: observables belonging to different sites have zero brackets. It does not assert that the $N\times N$ matrices at distinct sites commute under matrix multiplication.

Fix distinct numbers z_a and diagonal constant matrices

$$C=\mathrm{diag}(c_1,\dots,c_N),\qquad B=\mathrm{diag}(b_1,\dots,b_N).$$

Use the convention

$$T(u)=C\left(I+\frac{X^{(M)}}{u+z_M}\right)\cdots\left(I+\frac{X^{(1)}}{u+z_1}\right), \tag{5.6}$$

$$L(u)=B+\sum_{a=1}^{M}\frac{X^{(a)}}{u+z_a}. \tag{5.7}$$

The poles are $-z_a$. Relative to (4.16), this replaces the earlier pole parameters by their negatives. The product in (5.6) is ordered with the larger site label on the left. This order determines its second coefficient at infinity.

Each local factor satisfies (5.1), with the same $r=P_{12}/(u-v)$. Products of independent factors satisfy it too:

$$\begin{aligned}\big\{K_1J_1\overset{\otimes}{,}K_2J_2\big\}&=[r,K_1K_2]J_1J_2+K_1K_2[r,J_1J_2]\\&=[r,(KJ)_1(KJ)_2].\end{aligned}$$

Here only factors in different auxiliary slots were interchanged, and ultralocality removed the mixed brackets. Multiplication by C preserves the relation because $[P_{12},C\otimes C]=0$. For the sum, the entry brackets give

$$\big\{L_1(u)\overset{\otimes}{,}L_2(v)\big\}=\sum_a\frac{[P_{12},X_2^{(a)}]}{(u+z_a)(v+z_a)}=[r,L_1(u)+L_2(v)],$$

using $[P_{12},B_1+B_2]=0$. Thus both constructions provide commuting spectral coefficients.

For the multisite limit, replace every z_a by z_a/ε, evaluate at u/ε, and set $C_\varepsilon=I+\varepsilon B$. The product becomes

$$(I+\varepsilon B)\prod_{a=M}^{1}\left(I+\varepsilon\frac{X^{(a)}}{u+z_a}\right)=I+\varepsilon L(u)+O(\varepsilon^2).$$

Only the first-order sum survives; the factor order enters at higher orders.

5.3 Expanding a determinant at infinity

Two different variables play different roles. In $\det(tI + L(u))$, t is the characteristic variable and u is the spectral parameter. The determinant is a polynomial of degree N in t, but a rational function of u, with a generally infinite Laurent expansion at $u = \infty$.

For a matrix D invertible over a rational function field and $Y(u) = Y_1/u + Y_2/u^2 + O(u^{-3})$, factor out D:

$$\frac{\det(D + Y(u))}{\det D} = 1 + \frac{\operatorname{tr}(D^{-1}Y_1)}{u} + \frac{1}{u^2}\left(\operatorname{tr}(D^{-1}Y_2) + \frac{(\operatorname{tr}(D^{-1}Y_1))^2 - \operatorname{tr}((D^{-1}Y_1)^2)}{2}\right) + O(u^{-3}). \tag{5.8}$$

This follows from the first two elementary symmetric functions of $D^{-1}Y$. If $D = \operatorname{diag}(d_i)$, the quadratic term involving Y_1 is

$$\sum_{i<j} \frac{(Y_1)_{ii}(Y_1)_{jj} - (Y_1)_{ij}(Y_1)_{ji}}{d_i d_j}.$$

These identities hold formally in u^{-1}. Inverting D introduces denominators in t; multiplication by $\det D$ recovers the original polynomial in t.

5.4 One site: diagonal twists and partial fractions

First set $T(u) = C(I + X/u)$ and $L(u) = B + X/u$. For now assume the c_i are distinct and nonzero, and the b_i are distinct. Define

$$\det(tI + T(u)) = \det(tI + C)\sum_{j\geq 0} u^{-j}G^{(j)}(t),$$
$$\det(tI + L(u)) = \det(tI + B)\sum_{j\geq 0} u^{-j}F^{(j)}(t).$$

For one site the sums terminate at $j = N$. Set $\Delta_{ij} = x_{ii}x_{jj} - x_{ij}x_{ji}$ for $i \neq j$. Formula (5.8) gives

$$G^{(0)} = F^{(0)} = 1,$$
$$G^{(1)}(t) = \sum_i \frac{c_i x_{ii}}{t + c_i}, \qquad G^{(2)}(t) = \sum_{i<j} \frac{c_i c_j \Delta_{ij}}{(t + c_i)(t + c_j)},$$
$$F^{(1)}(t) = \sum_i \frac{x_{ii}}{t + b_i}, \qquad F^{(2)}(t) = \sum_{i<j} \frac{\Delta_{ij}}{(t + b_i)(t + b_j)}.$$

To separate the coefficients further, use

$$\frac{1}{(t+\alpha)(t+\beta)} = \frac{1}{\beta - \alpha}\left(\frac{1}{t+\alpha} - \frac{1}{t+\beta}\right). \tag{5.9}$$

Consequently

$$G^{(2)}(t) = \sum_i \frac{c_i}{t + c_i}\sum_{j\neq i} \frac{c_j(x_{ij}x_{ji} - x_{ii}x_{jj})}{c_i - c_j}, \tag{5.10}$$

$$F^{(2)}(t) = \sum_i \frac{1}{t + b_i}\sum_{j\neq i} \frac{x_{ij}x_{ji} - x_{ii}x_{jj}}{b_i - b_j}. \tag{5.11}$$

The numerator and denominator signs must be chosen together. For instance, the residue of $\Delta_{ij}/((t + b_i)(t + b_j))$ at $t = -b_i$ is $\Delta_{ij}/(b_j - b_i)$.

A two-by-two check. For $X = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$,

$$\det(tI + B + X/u) = (t+b_1)(t+b_2) + \frac{(t+b_2)a + (t+b_1)d}{u} + \frac{ad-bc}{u^2}.$$

Thus $F^{(2)} = (ad-bc)/((t+b_1)(t+b_2))$, fixing the signs in (5.11) directly. The analogous twisted determinant has $c_1c_2(ad-bc)$ as its coefficient of u^{-2}.

The two coefficient algebras are related when $B = C^{-1}$:

$$\det(tI + C(I + X/u)) = t^N \det C \, \det(C^{-1} + t^{-1}I + X/(ut)).$$

For arbitrary independently fixed B, C they need not be the same subalgebra. They need not always differ either: in size two, distinct diagonal twists give the same algebra generated by $a+d$, $ad-bc$, and a.

The corner degeneration. With $c_i = \delta^{i-1}$, extract the coefficient of t^{N-k} and normalize by $\delta^{-k(k-1)/2}$. The principal-minor calculation in Section 4.7 gives $\det(I_k + X_{(k)}/u)$ in the limit. Its coefficients are the Gelfand-Tsetlin generators. Scaling all c_i by one extra common factor δ only changes the scalar normalization, so $c_i = \delta^i$ gives the same limiting algebra. Individual simple-pole residues need not have finite limits when the c_i coalesce; normalized polynomial coefficients are the appropriate quantities to compare.

5.5 Several sites: the first two Laurent coefficients

For the partial-fraction formulas below, keep the assumptions that the b_i are pairwise distinct, the c_i are pairwise distinct, and $c_i \neq 0$. The determinant expansions themselves do not require distinct diagonal parameters. Put

$$\overline{X} = \sum_a X^{(a)}, \qquad Z_X = \sum_a z_a X^{(a)}, \qquad W = -Z_X + \sum_{a<b} X^{(b)}X^{(a)}.$$

Since $(u+z_a)^{-1} = u^{-1} - z_a u^{-2} + O(u^{-3})$, the ordered product and the sum expand as

$$T(u) = C\big(I + \overline{X}/u + W/u^2 + O(u^{-3})\big), \qquad L(u) = B + \overline{X}/u - Z_X/u^2 + O(u^{-3}).$$

Write $d_i = \overline{x}_{ii}$ and $\overline{\Delta}_{ij} = d_i d_j - \overline{x}_{ij}\overline{x}_{ji}$. Then

$$G^{(1)}(t) = \sum_i \frac{c_i d_i}{t+c_i}, \qquad F^{(1)}(t) = \sum_i \frac{d_i}{t+b_i},$$

$$G^{(2)}(t) = \sum_i \frac{c_i W_{ii}}{t+c_i} + \sum_{i<j} \frac{c_i c_j \overline{\Delta}_{ij}}{(t+c_i)(t+c_j)}, \tag{5.12}$$

$$F^{(2)}(t) = -\sum_i \frac{(Z_X)_{ii}}{t+b_i} + \sum_{i<j} \frac{\overline{\Delta}_{ij}}{(t+b_i)(t+b_j)}. \tag{5.13}$$

In particular,

$$W_{ii} = -\sum_a z_a x_{ii}^{(a)} + \sum_{a<b}\sum_j x_{ij}^{(b)} x_{ji}^{(a)}.$$

Reversing the local factor order replaces $X^{(b)}X^{(a)}$ by $X^{(a)}X^{(b)}$ here; ultralocality does not remove this difference.

Partial fractions now give

$$G^{(2)}(t) = \sum_i \frac{c_i R_i^{\mathrm{tr}}}{t + c_i}, \qquad R_i^{\mathrm{tr}} = W_{ii} + \sum_{j \neq i} \frac{c_j(\overline{x}_{ij}\overline{x}_{ji} - d_i d_j)}{c_i - c_j},$$

$$F^{(2)}(t) = \sum_i \frac{R_i^{\mathrm{rat}}}{t + b_i}, \qquad R_i^{\mathrm{rat}} = -(Z_X)_{ii} + \sum_{j \neq i} \frac{\overline{x}_{ij}\overline{x}_{ji} - d_i d_j}{b_i - b_j}.$$

Residues in the scalar parameter t are linear combinations of the commuting spectral coefficients. Hence the R_i commute within each fixed family.

Diagonal terms and dynamical Hamiltonians. The first coefficients show that each d_i belongs to the same commuting algebra when the diagonal parameters are distinct (and $c_i \neq 0$). One may therefore replace the residues by

$$D_i^{\mathrm{tr}} = W_{ii} + \sum_{j \neq i} \frac{c_j \overline{x}_{ij}\overline{x}_{ji}}{c_i - c_j},$$

$$D_i^{\mathrm{rat}} = -(Z_X)_{ii} + \sum_{j \neq i} \frac{\overline{x}_{ij}\overline{x}_{ji}}{b_i - b_j}. \tag{5.14}$$

The change adds an explicit polynomial in the commuting diagonal observables d_i. Thus the D_i^{tr} commute with each other, as do the D_i^{rat}. These are classical dynamical Hamiltonians in the present parameter convention. Their relation to Gaudin Hamiltonians will become concrete in the rank-one realization below.

The rational or trigonometric distinction here concerns the diagonal parameters, rather than the auxiliary r-matrix alone. If $c_i = e^{\lambda_i}$, then

$$\frac{c_j}{c_i - c_j} = \frac{1}{e^{\lambda_i - \lambda_j} - 1} = \frac{1}{2}\left(\coth\frac{\lambda_i - \lambda_j}{2} - 1\right),$$

whereas the rational family has $1/(b_i - b_j)$. The quantum counterparts occur in the duality between KZ and dynamical operators [10, 11]. Passing to quantum operators requires an ordering prescription and may introduce lower-degree terms; classical commutativity alone does not prove commutativity of an arbitrary ordered replacement. In particular, the diagonal-product terms in the determinant residues should be retained until the chosen normalization has been specified.

5.6 The Gaudin Hamiltonians from the characteristic polynomial

Now expand in t, not in u^{-1}:

$$\det(tI + L(u)) = \sum_{k=0}^{N} t^{N-k} H^{(k)}(u), \qquad H^{(0)} = 1. \tag{5.15}$$

The functions $H^{(k)}(u)$ Poisson-commute for all u. The first two are

$$H^{(1)}(u) = \operatorname{tr} L(u), \qquad H^{(2)}(u) = \frac{(\operatorname{tr} L(u))^2 - \operatorname{tr} L(u)^2}{2}.$$

Set $s_a = \operatorname{tr} X^{(a)}$, $\kappa_a = \frac{1}{2}\operatorname{tr}((X^{(a)})^2)$, and $\Omega_{ab} = \operatorname{tr}(X^{(a)}X^{(b)}) = \Omega_{ba}$. Direct expansion gives

$$H^{(1)}(u) = \operatorname{tr} B + \sum_a \frac{s_a}{u+z_a},$$
$$\frac{1}{2}\operatorname{tr} L(u)^2 = \frac{1}{2}\operatorname{tr} B^2 + \sum_a \frac{\kappa_a}{(u+z_a)^2} + \sum_a \frac{\mathcal{H}_a}{u+z_a}. \tag{5.16}$$

Here the Gaudin Hamiltonians are

$$\boxed{\mathcal{H}_a = \operatorname{tr}(BX^{(a)}) - \sum_{b\neq a} \frac{\Omega_{ab}}{z_a - z_b}.} \tag{5.17}$$

To obtain the simple-pole term, use

$$\frac{1}{(u+z_a)(u+z_b)} = \frac{1}{z_b - z_a}\left(\frac{1}{u+z_a} - \frac{1}{u+z_b}\right).$$

The minus sign in (5.17) comes from placing the poles at $-z_a$. With poles at ζ_a, where $\zeta_a = -z_a$, it becomes the usual plus sign in $\Omega_{ab}/(\zeta_a - \zeta_b)$.

For the characteristic coefficient itself, the full result is

$$H^{(2)}(u) = e_2(B) + \sum_a \frac{\frac{1}{2}s_a^2 - \kappa_a}{(u+z_a)^2} + \sum_a \frac{J_a}{u+z_a}, \tag{5.18}$$
$$J_a = (\operatorname{tr} B)s_a - \operatorname{tr}(BX^{(a)}) - \sum_{b\neq a} \frac{s_a s_b - \Omega_{ab}}{z_a - z_b}$$
$$= (\operatorname{tr} B)s_a - \mathcal{H}_a - \sum_{b\neq a} \frac{s_a s_b}{z_a - z_b}. \tag{5.19}$$

For example $e_2(B) = \sum_{i<j} b_i b_j$. Formula (5.18) includes the double poles even when they disappear in a particular representation. They are the second elementary invariants $e_2(X^{(a)})$ of the site matrices.

The s_a and κ_a are site Casimirs in the matrix Poisson algebra. Equation (5.19) says that J_a equals $-\mathcal{H}_a$ plus a function of these Casimirs. Thus, on a product of fixed coadjoint orbits, the Hamiltonian vector fields satisfy $X_{J_a} = -X_{\mathcal{H}_a}$: they generate the same trajectories with reversed time. The sign is part of the relation. Since the J_a belong to the commuting spectral family, it follows that

$$\{\mathcal{H}_a, \mathcal{H}_b\} = 0, \qquad \sum_a \mathcal{H}_a = \operatorname{tr}(B\overline{X}).$$

The second identity follows by pairing the (a,b) and (b,a) interaction terms. For $B = 0$ their sum is zero. The letter $\mathcal{H}$ here denotes a classical observable, not a quantum operator.

Two sites. For $M = 2$, write $\Omega = \operatorname{tr}(X^{(1)}X^{(2)})$. Then

$$\mathcal{H}_1 = \operatorname{tr}(BX^{(1)}) - \frac{\Omega}{z_1 - z_2}, \qquad \mathcal{H}_2 = \operatorname{tr}(BX^{(2)}) + \frac{\Omega}{z_1 - z_2}.$$

Their sum is $\operatorname{tr}(B(X^{(1)} + X^{(2)}))$; for $B = 0$ they are negatives of each other. Higher characteristic coefficients supply additional commuting observables when $N > 2$. Finding a commuting family and proving that it is complete are separate tasks.

5.7 Hamiltonian evolution and a Lax equation

For the flow generated by $\mathcal{H}_a$, the trace-gradient formula (3.6) gives

$$\{\mathcal{H}_a, X^{(a)}\} = \left[B - \sum_{b\neq a} \frac{X^{(b)}}{z_a - z_b}, X^{(a)}\right],$$
$$\{\mathcal{H}_a, X^{(b)}\} = \left[-\frac{X^{(a)}}{z_a - z_b}, X^{(b)}\right] \quad (b \neq a).$$

The site eigenvalues are preserved because these are commutator flows. Define $M_a(u) = -X^{(a)}/(u+z_a)$. Then

$$\frac{dL(u)}{d\tau_a} = \{\mathcal{H}_a, L(u)\} = [M_a(u), L(u)]. \tag{5.20}$$

Indeed the B term is $[B, X^{(a)}]/(u+z_a)$, and each mixed term uses the same partial-fraction identity as above. Taking a trace of any power of L gives zero derivative. Thus the Hamiltonian description and the isospectral Lax description agree.

5.8 Exercises

Take canonical coordinates $p_i^{(a)}, q_i^{(a)}$, with $i = 1, \ldots, N$ and $a = 1, \ldots, M$, satisfying

$$\{p_i^{(a)}, q_j^{(b)}\} = \delta_{ij}\delta_{ab}, \qquad \{p_i^{(a)}, p_j^{(b)}\} = \{q_i^{(a)}, q_j^{(b)}\} = 0.$$

Set $x_{ij}^{(a)} = q_i^{(a)} p_j^{(a)}$ and collect the coordinates into

$$Q = (q_i^{(a)})_{i,a} \in \mathrm{Mat}_{N\times M}, \qquad P = (p_i^{(a)})_{a,i} \in \mathrm{Mat}_{M\times N}\,.$$

Here P is the rectangular matrix of momenta; the permutation of auxiliary tensor factors is written P_{12}. Keep the diagonal matrices B, C and the parameters z_a from (5.6) and (5.7), and put

$$Z = \mathrm{diag}(z_1, \ldots, z_M), \qquad A = tI_N + B, \qquad D = uI_M + Z.$$

Work first where A, D are invertible, or over the rational function field in t, u.

(a) **Rank-one realization.** Prove that the functions $x_{ij}^{(a)}$ satisfy

$$\{x_{ij}^{(a)}, x_{kl}^{(b)}\} = \delta_{ab}(\delta_{jk}x_{il}^{(a)} - \delta_{il}x_{kj}^{(a)}).$$

Determine how this bracket changes if one instead sets $x_{ij}^{(a)} = p_i^{(a)} q_j^{(a)}$.

(b) **Determinant identity.** Prove the equalities

$$\det(I_N + A^{-1}QD^{-1}P) = \det(I_M + D^{-1}PA^{-1}Q)$$
$$= \frac{\det\begin{pmatrix} A & Q \\ -P & D \end{pmatrix}}{\det A \,\det D}.$$

Explain why the resulting denominator-cleared identity extends beyond the invertible locus.

(c) **Dual spectral matrices.** Using part (b), find an $M \times M$ matrix giving an equivalent characteristic-determinant description of the $N \times N$ matrix

$$L(u) = B + \sum_a \frac{X^{(a)}}{u + z_a}.$$

Specify the scalar prefactors, identify the exchanged matrix indices and sites, and check the Poisson-bracket sign of the dual residues. Find an analogous $M \times M$ spectral description for the ordered product $T(u)$ in (5.6), retaining its factor order.

The solutions below treat the three parts in this order.

5.9 Solution to (a): a canonical rank-one realization

Take canonical variables $p_i^{(a)}, q_i^{(a)}$, with

$$\{p_i^{(a)}, q_j^{(b)}\} = \delta_{ij}\delta_{ab}, \qquad \{p_i^{(a)}, p_j^{(b)}\} = \{q_i^{(a)}, q_j^{(b)}\} = 0.$$

The realization compatible with (5.5) is

$$\boxed{x_{ij}^{(a)} = q_i^{(a)} p_j^{(a)}.} \tag{5.21}$$

Using the canonical bracket and the Leibniz rule,

$$\begin{aligned}\{q_i^{(a)} p_j^{(a)}, q_k^{(b)} p_l^{(b)}\} &= \delta_{ab}\left(q_i^{(a)}\{p_j^{(a)}, q_k^{(a)}\}p_l^{(a)} + q_k^{(a)}\{q_i^{(a)}, p_l^{(a)}\}p_j^{(a)}\right) \\ &= \delta_{ab}(\delta_{jk} q_i^{(a)} p_l^{(a)} - \delta_{il} q_k^{(a)} p_j^{(a)}).\end{aligned}$$

This is precisely the required matrix-entry bracket. By contrast, $p_i^{(a)} q_j^{(a)}$ gives its negative. Since the variables commute under multiplication, (5.21) can equally be written $p_j^{(a)} q_i^{(a)}$.

In vector notation $X^{(a)} = q^{(a)}(p^{(a)})^{\mathsf{T}}$, so its rank is at most one and

$$(X^{(a)})^2 = s_a X^{(a)}, \qquad s_a = (p^{(a)})^{\mathsf{T}} q^{(a)}, \qquad e_2(X^{(a)}) = 0.$$

Thus all double poles in (5.18) vanish in this realization. The pullback of s_a commutes with all matrix-entry observables $x_{ij}^{(b)}$, but it is not central in the full canonical algebra:

$$\{s_a, q_i^{(b)}\} = \delta_{ab} q_i^{(b)}, \qquad \{s_a, p_i^{(b)}\} = -\delta_{ab} p_i^{(b)}.$$

Its Hamiltonian flow generates the rescaling of $q^{(a)}, p^{(a)}$. Thus the ambient algebra matters when calling an observable a Casimir. The map from the $2N$ canonical coordinates at one site is not a parametrization of all $N \times N$ matrices. It has the rescaling redundancy $(q, p) \mapsto (\lambda q, \lambda^{-1} p)$ for $\lambda \in \mathbb{C}^\times$ and its image lies in the rank-at-most-one locus. For $s_a \neq 0$, fixing s_a and taking the quotient by this rescaling yields the rank-one orbit of complex dimension $2N-2$. Its half-dimension is $N - 1$.

The interaction becomes

$$\Omega_{ab} = \big((p^{(a)})^{\mathsf{T}} q^{(b)}\big)\big((p^{(b)})^{\mathsf{T}} q^{(a)}\big).$$

This gives an explicit polynomial-rational expression for every $\mathcal{H}_a$ in canonical variables.

5.10 Solution to (b): the block determinant identity

Collect the vectors into rectangular matrices

$$Q = (q_i^{(a)})_{i,a} \in \mathrm{Mat}_{N\times M}, \qquad P = (p_i^{(a)})_{a,i} \in \mathrm{Mat}_{M\times N}, \qquad Z = \mathrm{diag}(z_1, \dots, z_M).$$

Here P is a rectangular matrix of canonical variables, distinct from the permutation P_{12}. Then

$$L(u) = B + Q(uI_M + Z)^{-1}P.$$

Put $A = tI_N + B$ and $D = uI_M + Z$. The determinant identity is

$$\boxed{\det(I_N + A^{-1}QD^{-1}P) = \det(I_M + D^{-1}PA^{-1}Q).} \tag{5.22}$$

It compares determinants of different sizes. To prove it, consider

$$\mathcal{K}(t,u) = \begin{pmatrix} A & Q \\ -P & D \end{pmatrix}.$$

Elimination using the invertible lower-right block gives

$$\det \mathcal{K} = \det D \, \det(A + QD^{-1}P).$$

Eliminating the upper-left block instead gives

$$\det \mathcal{K} = \det A \, \det(D + PA^{-1}Q).$$

Dividing by $\det A \det D$ proves (5.22). This is the Schur-complement proof of Sylvester's determinant identity. For a block matrix with a *plus* P in the lower-left block, both Schur complements instead contain a minus sign. The choice $-P$ above keeps the plus signs in $L(u)$.

Equivalently,

$$\det(uI_M + Z)\det(tI_N + L(u)) = \det(tI_N + B)\det(uI_M + \widetilde{L}(t)), \tag{5.23}$$

where $\widetilde{L}(t) = Z + P(tI_N + B)^{-1}Q$. After multiplying out and cancelling the scalar denominators, both sides equal the polynomial $\det \mathcal{K}(t,u)$. This polynomial identity extends to the zeros of $\det A \det D$; the individual rational Lax matrices still have their stated poles. Here t is the spectral parameter of the dual matrix and u its characteristic variable, so their original roles are exchanged. It is the elementary determinant mechanism behind rank-size spectral duality [8, Section 3.1]; the dual moment-map construction is developed in [9].

5.11 Solution to (c): the dual linear and quadratic matrices

The residues of the $M \times M$ matrix $\widetilde{L}(t)$ are

$$\widetilde{L}(t) = Z + \sum_{i=1}^{N} \frac{Y^{(i)}}{t + b_i}, \qquad Y_{ab}^{(i)} = p_i^{(a)} q_i^{(b)}.$$

A direct canonical-bracket computation gives

$$\{Y_{ab}^{(i)}, Y_{cd}^{(j)}\} = \delta_{ij}(\delta_{ad}Y_{cb}^{(i)} - \delta_{bc}Y_{ad}^{(i)}).$$

Thus these residues carry the opposite sign of the Lie-Poisson bracket. To use exactly the earlier sign convention, transpose the matrix:

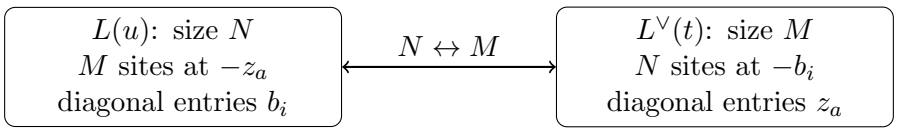
$$L^\vee(t) = \widetilde{L}(t)^{\mathsf{T}} = Z + \sum_i \frac{X^{\vee(i)}}{t+b_i},$$
$$X^{\vee(i)}_{ab} = q_i^{(a)} p_i^{(b)}, \qquad \{X^{\vee(i)}_{ab}, X^{\vee(j)}_{cd}\} = \delta_{ij}(\delta_{bc} X^{\vee(i)}_{ad} - \delta_{ad} X^{\vee(i)}_{cb}). \tag{5.24}$$

Determinants are unchanged by transposition, so (5.23) remains valid with $L^\vee$. It is a Gaudin matrix of size M with N sites. The two descriptions use the same canonical phase space; independent sites refer to distinct site labels within either description, not to independence between the original and dual matrices. The dual matrix has z_a as its diagonal constant entries and $-b_i$ as its poles. The original matrix had size N, M sites, diagonal constants b_i, and poles $-z_a$.

Its quadratic Gaudin Hamiltonians are

$$\begin{aligned}\mathcal{H}_i^\vee &= \operatorname{tr}(Z X^{\vee(i)}) - \sum_{j\neq i} \frac{\operatorname{tr}(X^{\vee(i)} X^{\vee(j)})}{b_i - b_j} \\ &= \sum_a z_a x_{ii}^{(a)} - \sum_{j\neq i} \frac{\overline{x}_{ij}\overline{x}_{ji}}{b_i - b_j} = -D_i^{\mathrm{rat}}.\end{aligned}$$

The last equality proves, in canonical coordinates, how the rational dynamical Hamiltonians become Gaudin Hamiltonians after exchanging the two sets of indices. The sign reflects our choices $u + z_a$ and $t + b_i$.



one block determinant $\det \mathcal{K}(t,u)$

A quadratic construction on the dual sites. The same dual residues also give a multiplicative matrix

$$T^\vee(t) = C^\vee \left(I_M + \frac{X^{\vee(N)}}{t+b_N}\right) \cdots \left(I_M + \frac{X^{\vee(1)}}{t+b_1}\right),$$

for any fixed $M \times M$ twist $C^\vee$. Ultralocality in i proves the quadratic bracket, just as in (5.6). This supplies dual-size versions of both L and T. The block determinant above establishes equality of the *linear* spectral families; it does not assert an identical determinant relation for arbitrary products T and $T^\vee$. That stronger multiplicative duality requires additional data.

An exact spectral representation of the ordered product. For the rank-one product (5.6), the additional data can be written explicitly. Set $A_0 = PQ$ and let S be its strictly lower triangular part in the site indices:

$$S_{ab} = \begin{cases} (p^{(a)})^{\mathsf{T}} q^{(b)}, & a > b, \\ 0, & a \leq b. \end{cases}$$

Then the product identity is

$$\prod_{a=M}^{1} \left(I_N + \frac{q^{(a)}(p^{(a)})^{\mathsf{T}}}{u+z_a}\right) = I_N + Q(uI_M + Z - S)^{-1} P. \tag{5.25}$$

To prove it, write $D = uI_M + Z$. Since $D^{-1}S$ is strictly lower triangular, it is nilpotent and

$$(D-S)^{-1} = \sum_{k=0}^{M-1} (D^{-1}S)^k D^{-1}.$$

For example, the term with one S is

$$QD^{-1}SD^{-1}P = \sum_{a>b} \frac{q^{(a)}(p^{(a)})^{\mathsf{T}} q^{(b)}(p^{(b)})^{\mathsf{T}}}{(u+z_a)(u+z_b)}.$$

Higher terms run over strictly decreasing chains of site labels, exactly the terms in the expanded ordered product. This proves the identity.

Apply the Schur-complement argument to

$$\begin{pmatrix} tI_N + C & CQ \\ -P & uI_M + Z - S \end{pmatrix}.$$

Since $Z - S$ is lower triangular with diagonal z_a, the result is

$$\det(uI_M + Z)\det(tI_N + T(u)) = \det(tI_N + C)\det(uI_M + K_C(t)), \tag{5.26}$$

where

$$K_C(t) = Z - S + P(tI_N + C)^{-1}CQ = Z - S + \sum_i \frac{c_i Y^{(i)}}{t + c_i}.$$

This is an exact dual-size spectral matrix for the original product. Its variable-dependent triangular term $-S$ distinguishes it from the linear Gaudin matrix. It cannot simply be treated as a constant twist plus independent residues with the original linear bracket. Nevertheless its spectral coefficients commute: after clearing the scalar denominators, (5.26) identifies them with the commuting coefficients of $\det(tI + T(u))$. The derivation explains why the site order must be preserved in a multiplicative duality.

A one-site check of the spectral identity. Let $M = 1$, $z = z_1$, and $X = qp^{\mathsf{T}}$. Then

$$\det(tI_N + B + X/(u+z)) = \det(tI_N + B)\left(1 + \frac{p^{\mathsf{T}}(tI_N + B)^{-1}q}{u+z}\right).$$

Multiplying by $u + z$ gives

$$(u+z)\det(tI_N + L(u)) = \det(tI_N + B)\left(u + z + \sum_i \frac{p_i q_i}{t + b_i}\right).$$

The right factor is the characteristic polynomial of a scalar dual Lax matrix with N poles. This verifies both the scalar prefactors and the plus signs in the general block formula.

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